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Modeling and Discrete Simulation

Subject 2

Simulation Examples

Mujdat Soyturk, Ph.D.
Associate Professor

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Slides are derived from the course textbook "Banks, Carson, Nelson, Nicol. Discrete-Event System Simulation, 5th Edition, Prentice Hall, 2010".

How to simulate randomness?

- DES models contain one or more random variables.
  - Coin tossing; either head or tail.
  - Queuing models: both service times and interarrival times (from a statistical distribution)
- Random numbers are generated between 0 and 1.
- Random variable: randomly generated quantity with a specified statistical distribution.

Random Number Generation

- Random numbers are generated by Random Number Generators (RNG)
- generated in sequence
  - numbers should be uniformly distributed between 0 and 1.
  - Subsequent numbers generated should be statistically independent of all previous numbers generated.
- Actually, generated numbers are pseudo-random numbers.
  - We want to keep control on the simulation development

  ➔ reproduce the sequence at any time.
    - Helps in development and debugging

Example 1: Coin Tossing

- An example to Monte Carlo simulation. There is no events or clock times being tracked.

Example 2: Random Service Time

- It shows how to generate random samples from arbitrary discrete distribution.
- The key principal, the transformation from a random number to a service time of a prescribed probability, easily generalizes to any discrete distribution with a finite number of possible values.

Example 2: Random Service Time

	A	B	C	D
4		Service Time	Probability	Cumulative Probability
5				
6				
7		3	0.30	0.30
8		6	0.45	0.75
9		10	0.25	1.00
10				
11		Number of Callers=		25
12		Simulation Table		
13		Step	Activity	
14				
15		Caller	Service Time	
16		1	6	
17		2	6	
18		3	10	
19		4	6	

Example 3: Random Arrival Times

- To show
  - how to generate random interarrival times (times between successive arrivals)
  - To compute arrival times of customers or other entities to a system
- Interarrival times (and service times) are examples of simulation-generated time durations, called activity times or activities.
- Interarrival times (and service times) are distinguished from arrival times, which are times on a simulation clock representing the time-of-arrival to a facility.

Example 3: Random Arrival Times

- A simulation clock is a key component of every dynamic discrete-event simulation.
- The example 3 is a dynamic, event-based model.
  - Dynamic simply means time-based, with system state changing over-time.
  - Dynamic, event-based means that the model tracks the progression of event occurrences over time.

Some guidelines

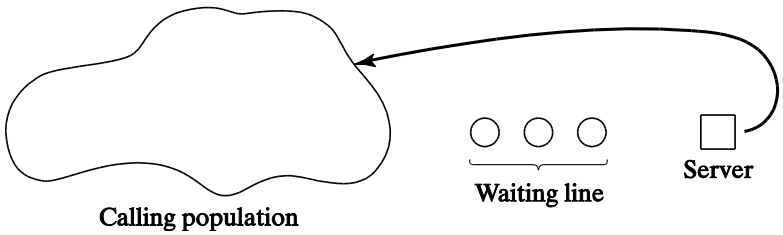
- The model developer must define the model inputs, the system states, and the model outputs.
  - Inputs are the exogenous variables that are (usually) defined independently of other system characteristics.
    - Probability of a head on a coin toss.
    - The service time and interarrival time distributions in a queuing system.
    - Demand distribution in an inventory model.
  - Other inputs, such as policy or decision parameters may be constants.

Some guidelines

- Outputs are used to compute measures of a system performance (also known as model responses or simply responses).
  - Output Ex. An individual customer's waiting time in a queue,
  - Response Ex. Average waiting time in a queue.
  - Output Ex. The cost of an individual transaction in an inventory system.
  - Response Ex. Average cost per unit time in an inventory system.
- Outputs are the results produced with the execution of the simulation.
- The responses are computed afterwards.

Example 5
Single Server Queue

- A queuing system is described by
  - Its calling population
  - The nature of arrivals
  - The service mechanism
  - The system capacity
  - The queuing discipline



Example 5
Single Server Queue

- In the example:
  - Unlimited potential calling population
  - Constant arrival rate
  - Random length service times (from a distribution)
  - Unlimited system capacity (unlimited queue size)
  - FIFO (FCFS)

- Overall effective arrival rate must be less than the total service time.

Example 5
Single Server Queue

- The state of the system is
  - the number of units in the system and
  - the status of the server, busy or idle.
- An event is a set of circumstances that causes an instantaneous change in the state of the system.
  - In a single-channel queuing system,
    - two possible events
      - the entry of a unit into the system (the arrival event)
      - the completion of a service on a unit (the departure event).
- The queuing system includes the server, the unit being served (if one is being serviced), and the units in the queue (if any are waiting). The simulation clock is used to track simulated time.